HKE's PDA COLLEGE OF ENGINEERING (AUTONOMOUS), KALABURAGI- KARNATAKA

Department of Electronics and Communication Engineering

Introduction to Electronics and Communication Engineering

[Quote : " Don't allow the fear of failure, to override the joy of learning " .]

Faculty : S.K.Aurad

Module - 1 : Power Supply and Amplifiers

1. Define the term 'Electronics'. Mention the value of charge and mass of an electron.
2. Draw and explain the block diagram of DC power supply. Sketch the signals/waveforms at each stage.
3. With a neat figure and waveforms, explain the working of Half-wave rectifier. Mention important equations.
4. With a neat figure and waveforms, explain the working of centre-tapped FWR. Mention important equations.
5. With a neat figure and waveforms, explain the working of Bridge rectifier. Mention important equations.
6. Compare HWR, c-t-FWR, and Bridge Rectifiers.
7. Derive the expression for Ripple factor, Rectification efficiency of HWR.
8. Derive the expression for Ripple factor, Rectification efficiency of FWR.
9. What are Filters ? With relevant circuit diagram and waveforms, explain the working of HWR with Capacitor filter. Mention the important equations.
10. With relevant circuit diagram and waveforms, explain the working of FWR with Capacitor filter. Mention the important equations.
11. What is a Zener Diode ? What is Zener Breakdown ? Sketch the symbol and it's V-I Characteristics and label important points.
12. With a neat circuit diagram, explain the working of Voltage Doubler.
13. What are Amplifiers ? List and explain different types of Amplifiers. Sketch and label the frequency response of various Amplifiers.
14. Define the following terms: Gain, Input resistance and output resistance of CE Transistor Amplifier.
15. Draw the circuit of two-stage RC coupled Amplifier. Sketch and label it's frequency response.
16. With a simple circuit, discuss Zener diode as Voltage Regulator, write the relevant design equations.
17. An AC supply of 230V is applied to HWR through a transformer of turns ratio 10:1. Find a) dc output voltage and b) Peak Inverse Voltage(PIV). Assume ideal diode.
18. In a HWR the input voltage is 230V and transformer ratio is 2:1. Determine the maximum and Average values of power delivered to the load. Assume R_L=200Ω.
19. A FWR has a load of 1kΩ. The AC voltage applied is 200-0-200V. Calculate a) dc voltage b) dc current c) ripple voltage. Assume ideal diodes. Draw the circuit.
20. In a c-t FWR, R_L is 2kΩ. Each diode has forward resistance of 10Ω. The peak value of voltage across secondary is 210V. Find the peak value of current, rms value of current, and ripple factor. Draw the circuit.
21. An AC supply of 230V is applied to a FWR through a transformer of turns-ratio 2:1 to a resistive load 10kΩ. Compute a) dc output power, b)ac input power, c) efficiency. Assume ideal diode.
22. For a Zener Regulator, if V_z=10V, R_s=1kΩ, R_L=2kΩ, and the input voltage varies from 22V to 40V. find the maximum and minimum values of Zener current. Draw the circuit.
23. In a Zener Regulator, the input DC is 10V with 20% variations. The output requirements are 5V, 20mA. Assume $I_z$min, $I_z$max as 5mA and 80mA. Design the Zener Regulator. Draw the circuit.

Module - 2 : Oscillators and Operational Amplifiers (OpAmps)

24. Explain the concept of feedback with a neat figure. Define and Compare Negative and Positive feedback.
25. State Barkhausen Criteria. Mention the condition for Oscillations.
26. With a neat figure, equations and waveforms, explain the working of RC Phase Shift Oscillator.

27. With a neat figure, equations and waveforms, explain the working of Hartley Oscillator.
28. With a neat figure, equations and waveforms, explain the working of Colpitt's Oscillator.
29. What is Piezo-electric effect. With neat circuits and equations, explain the working of Crystal oscillator.
30. A RC phase shift oscillator is used to generate oscillations. If R=10kΩ, C=0.1uF and the gain of the amplifier used is 1000. Calculate the frequency of oscillations.
31. A Hartley Oscillator has L1=0.4mH, L2=0.2mH, M=0.1mH. Draw the circuit and determine the frequency of oscillations.
32. A Colpitt's Oscillator has C1=0.4mH, C2=0.2mH. Draw the circuit and determine the frequency of oscillations.
33. A Crystal has L=0.4H, C=0.085pF, C_M=1pF, R=5kΩ. Determine the series and parallel resonant frequency, Q factor of crystal.
34. What is an OPAMP ? Why is it called so ? What is Differential Amplifier ? Sketch its symbol and mention important details.
35. Draw the block diagram of OpAmp and explain the role of each block.
36. List and define various Opamp parameters.
37. Define the terms : Open-loop gain, Differential gain, Common-mode gain, input impedance, output impedance, CMRR, Slewrate, Bandwidth, PSRR, input offset voltage, input offset current.
38. List and write a note on characteristics of an Ideal OpAmp.
39. Write a note on Virtual Ground, explain using an opamp based circuit diagram.
40. With a neat circuit diagram using OpAmp, derivation, waveforms, explain the operation of Inverting Amplifier.
41. With a neat circuit diagram using OpAmp, derivation, waveforms, explain the operation of Unity Gain Amplifier.
42. With a neat circuit diagram using OpAmp, derivation, waveforms, explain the operation of Non-Inverting Amplifier.
43. With a neat circuit diagram using OpAmp, derivation, and waveforms, explain the operation of Voltage Follower.
44. With a neat circuit diagram using OpAmp, derivation, and waveforms, explain the operation of opamp adder/summer.
45. With a neat circuit diagram using OpAmp, derivation, and waveforms, explain the operation of Differentiator.
46. With a neat circuit diagram using OpAmp, derivation, and waveforms, explain the operation of Integrator.
47. With a neat circuit diagram using OpAmp, derivation, and waveforms, explain the operation of Comparator.
48. A differential Amplifier has open-loop gain of 100. The input signals v_1 and v_2 are 3.25v and 3.15v. Compute difference voltage, common mode voltage and the output voltage.
49. In an OpampInverting Amplifier circuit, R_f=100kΩ, R_1=10kohms, v_1=1volt. Calculate input current, Gain, and output voltage. Draw the circuit.
50. Design an Opamp Amplifier with a gain of +9. Assume R_f=12kohms. Compute output voltage, if input is 1V.
51. In an Opamp Inverting Adder, v1=-1v, v2=+2v, and v3= +4v, Compute the output voltage, if R_1=10kΩ,R_2 =20kΩR_3 = 30kΩ, R_f = 100kΩ.
52. For an Opamp Inverting amplifier, R_1=1kΩ, R_f =1MΩ. Assume an Ideal Amplifier, determine voltage gain, input resistance, output resistance. Draw the circuit using opamp.
53. In an Opamp Differentiator, a sinusoidal voltage of peak value 5mV and frequency 2kHz is applied as input. Find the output if R_f =100kΩ and C=1uF. Draw the circuit and sketch the input and output waveforms.
54. A 5mV, 1kHz sinusoidal signal is applied as input to an Opamp Integrator with R=100kΩ, C=1μf. Compute the output voltage. Sketch input and output waveforms. Draw the circuit and sketch input and output waveforms.